

Adaptive Multi-Stage Behavioural Progression Architecture for Early Detection of SIP Toll Fraud

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Abstract

Voice over IP (VoIP) has become an essential communication technology for modern organizations around the world because it supports flexible, scalable, and cost-effective communication. However, this same openness has also increased exposure to security threats, especially SIP toll fraud, which can cause serious financial and operational damage. In many cases, such attacks do not begin with fraudulent calling immediately. Instead, they develop through a sequence of suspicious activities such as reconnaissance, registration probing, credential abuse, compromise, and abnormal outbound call behaviour. This project is important because it addresses a real global problem in communication security and aims to support earlier detection before major fraud losses occur.

This proposal presents an Adaptive Multi-Stage Behavioural Progression Architecture for Early Detection of SIP Toll Fraud. The proposed research introduces a stage-aware detection approach that observes SIP signalling behaviour, reconstructs communication patterns, and analyses behavioural indicators across monitored entities such as IP addresses, extensions, trunks, and destinations. The framework combines behavioural progression logic with lightweight machine learning in order to improve detection accuracy while maintaining interpretability and practical usability. In addition, the study considers two implementation paths: an in-process PBX-based approach and an external integrated detection service, making the work both academically relevant and practically deployable.

At the current level, the proposal has successfully established the research problem, research objectives, conceptual framework, and methodological direction needed for the study. The proposed objectives are clearly aligned with the planned implementation and evaluation strategy, particularly in identifying early fraud indicators, modelling attack progression, refining risk scoring, and assessing deployment feasibility. Although the full prototype implementation and experimental validation are still to be completed, the proposal has achieved a strong foundation for carrying out the research in a systematic and realistic VoIP environment.

Keywords: SIP security, VoIP fraud detection, toll fraud, behavioural progression modelling, early detection, CDR analytics, SIP dialog analysis, risk scoring.

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1. Introduction

This report presents the project titled **Adaptive Multi-Stage Behavioural Progression Architecture for Early Detection of SIP Toll Fraud**. The study focuses on a technical security problem in modern Voice over IP (VoIP) environments, where attackers may not move directly to fraudulent calling, but instead progress through multiple behavioural stages such as reconnaissance, registration probing, credential abuse, compromise, and abnormal outbound call activity. The subject of this report is therefore the design of an early detection architecture that can observe this progression using SIP signalling behaviour, entity-level activity patterns, and operational telemetry available in real VoIP deployments. In addition to proposing a detection model, the report also considers practical deployment paths so that the research is not limited only to theory, but can be evaluated in a way that is relevant to operational PBX environments.

The remainder of this proposal is structured as follows. Section 2 reviews the literature related to SIP security, VoIP fraud detection, behavioural anomaly detection, and PBX integration approaches. Section 3 presents the research objectives, research questions, and key study definitions. Section 4 explains the research design and methodology, including the testbed, sampling strategy, data collection methods, and analysis techniques. Section 5 discusses the significance and expected contributions of the study. Section 6 outlines the ethical considerations. Section 7 presents the timeline and work plan, while Section 8 summarizes the required resources and budget. Finally, Section 9 provides the references and Section 10 includes the appendices.

1.1. Background

Voice over IP has become a core communication technology across enterprise, carrier, and service-provider environments because it offers flexibility, scalability, and cost efficiency. At the signalling layer, Session Initiation Protocol (SIP) is widely used to create, modify, and terminate communication sessions [8]. In practical VoIP operation, SIP exchanges generate rich behavioural information through messages such as REGISTER, INVITE, ACK, BYE, authentication challenges, response codes, dialog identifiers, and transaction timing. In parallel, PBX platforms expose monitoring and integration points such as Call Detail Records (CDRs), SIP logs, event streams, and management interfaces, creating a technically useful foundation for traffic observation and security analysis [12], [13].

However, this same openness also creates a strong attack surface. SIP toll fraud is especially important because it can cause direct financial loss while remaining difficult to detect in its early stages. In many cases, the attack does not appear as one single event. Instead, it evolves through a sequence of behaviours such as endpoint discovery, repeated authentication failures, successful credential use, abnormal call setup, and bursts of unauthorized outbound traffic [9], [10], [13]. Traditional defensive practice often relies on threshold-based alarms, static rules, or post-event CDR review. These methods may detect abuse after the damage has already occurred, but they are weaker at recognizing the behavioural transition that happens before the fraud burst stage. This creates a clear need for a more technical and progression-aware detection approach that can combine protocol semantics with operational system visibility.

1.2. Research Problem

The central research problem is the lack of an early detection model for SIP toll fraud that can capture behavioural progression across multiple protocol stages while still remaining practical for deployment in real VoIP environments. Existing studies have shown the value of anomaly detection, event correlation, and fraud monitoring in VoIP systems [9], [10], [13]. However, an important gap remains in integrating these ideas into a single framework that can reconstruct SIP behaviour, recognize stage transitions, generate meaningful risk scores, and support implementation choices suitable for operational environments.

More specifically, current approaches do not sufficiently address how early indicators such as repeated REGISTER failures, suspicious 401/403/407 response patterns, rapid failure-to-success transitions, unusual destination diversity, and sudden outbound calling bursts can be correlated as part of one evolving fraud lifecycle.

Accordingly, this study focuses on designing a progression-aware early-detection architecture that can identify suspicious SIP behaviour before large-scale toll fraud occurs. The detailed research questions and hypotheses are presented in Section 3.

This problem is significant because it has both academic and practical value. From an academic perspective, it extends the discussion from isolated SIP anomaly detection toward stage-aware behavioural modelling and early-warning design. From a practical perspective, it aims to reduce financial exposure and improve response time by identifying suspicious activity before full fraud execution. The problem is also suitable for a master's-level project because it can be addressed through a prototype-based implementation, controlled SIP traffic generation, simulated attack scenarios, and evaluation using signalling traces, event streams, and sanitized CDR data.

1.3. Objectives

To address the research problem, the study will pursue the following specific objectives:

1. **To design a stage-aware SIP toll fraud detection architecture** that reconstructs SIP dialogs and captures behavioural evidence relevant to early fraud detection.
2. **To develop a behavioural progression engine** that tracks attack evolution per monitored entity, such as extension, IP address, trunk, or destination group.
3. **To integrate a lightweight machine learning model** to refine detection outcomes and support progression-based risk scoring.
4. **To evaluate detection effectiveness** using simulated attack scenarios and realistic VoIP traffic by measuring detection accuracy, false positives, and detection latency.
5. **To assess implementation feasibility and computational overhead** under the selected deployment options by measuring processing cost, resource usage, and operational responsiveness.

2. Literature Review

This section reviews the most relevant research streams related to SIP toll fraud, VoIP anomaly detection, and operational fraud monitoring. The literature is organized around four themes that are directly relevant to this study: **real-time fraud detection**, **stateful and protocol-aware SIP analysis**, **behavioural profiling using CDR and telecom interaction data**, and **machine-learning-based SIP attack detection**. The purpose of the review is not only to summarize prior findings, but also to identify where existing knowledge remains insufficient for early warning of SIP toll fraud in practical VoIP environments.

2.1. Review of Existing Literature

A first important stream of literature focuses on real-time fraud detection in VoIP systems. Manunza et al. proposed Kerberos, a real-time fraud detection architecture for IMS-enabled VoIP networks that processes charging and call-evolution events using complex event processing (CEP) [14]. A major strength of this work is that it moves fraud analysis away from purely offline CDR review and demonstrates that event-driven online monitoring can support remediation while a call is still ongoing. This is highly relevant because toll fraud often causes financial damage in a short period of time. However, Kerberos is primarily built around charging and service events rather than detailed SIP signalling reconstruction. As a result, it shows the feasibility of online fraud monitoring, but it does not explicitly model SIP-level behavioural progression from early reconnaissance to fraud execution.

A second stream of literature deals with stateful and protocol-aware SIP anomaly detection. Seo et al. introduced SIPAD, which uses a stateful rule tree for SIP–VoIP anomaly detection [15]. Their work demonstrates that protocol-aware inspection improves coverage and reduces overhead compared with simpler stateless inspection methods. Likewise, Golait and Hubballi model anomalous VoIP behaviour using a discrete event system, showing that SIP behaviour can be represented as structured state transitions rather than as isolated packets or independent counters [17]. These studies are important because they justify the use of state, sequence, and protocol semantics in VoIP security monitoring. However, their primary emphasis is on anomaly detection and signalling irregularities in general rather than specifically on multi-stage toll fraud. In other words, they provide a strong methodological foundation for state-based detection, but do not fully operationalize toll fraud as a progression from scan to compromise to burst calling.

A third research direction focuses on behavioural profiling using CDRs and telecom interaction patterns. Chiappetta et al. present an anomaly-based analysis of the social behaviour of VoIP users using large-scale CDR data [16]. Their work is valuable because it shows how user-level calling behaviour, clustering, and relational patterns can reveal suspicious activity in operator environments. More recent telecom fraud research has gone further by applying graph-based models to fraud detection from large real-world calling datasets. For example, Hu et al. propose BTG, a graph-learning framework for telecom fraud detection [19], while Ren et al. develop a dynamic graph neural network model to identify collaborative fraudsters [20]. These studies show that behavioural context, temporal relations, and entity-to-entity structure are highly informative for fraud analytics. Nevertheless, these approaches are mainly CDR-centric or call-

log-centric, meaning they operate after or around completed communication events rather than directly at SIP signalling time. Therefore, while they are useful for modelling normal behaviour and supporting baselines, they are less suitable for immediate operational response inside a PBX or SIP environment.

A fourth stream concerns machine-learning-based SIP attack detection, particularly for SIP flooding and denial-of-service scenarios. Younes proposed a hybrid deep learning and entropy-based method for detecting SIP DDoS flooding attacks, including INVITE, REGISTER, and ACK flooding [18]. Similarly, Dassouki et al. proposed protection against cloud-based SIP flooding attacks using temporal and structural fingerprints [21]. These studies are technically important because they show that timing features, message fingerprints, entropy, and temporal learning can improve SIP attack detection accuracy. They also highlight that SIP message types and protocol-specific temporal dynamics provide meaningful inputs for intelligent detection models. However, their focus remains mainly on flooding and denial-of-service behaviour. They do not explicitly address toll fraud as a behavioural chain involving reconnaissance, authentication abuse, compromise, and high-cost outbound calling.

Taken together, the literature confirms several important points. First, real-time monitoring is feasible and necessary in VoIP fraud contexts [14]. Second, stateful and protocol-aware modelling provides clear advantages over purely stateless threshold-based approaches [15], [17]. Third, behavioural baselining using CDRs or telecom graphs is useful for understanding normal versus abnormal entity behaviour [16], [19], [20]. Fourth, machine learning can improve SIP attack classification, especially when temporal and protocol-specific features are included [18], [21]. However, the evidence is still fragmented across different problem settings. Prior work tends to study either real-time fraud events, SIP anomalies, CDR-based fraud analytics, or flooding attacks separately. There is limited integration of these strands into one architecture designed specifically for early detection of SIP toll fraud progression, as shown in Table 2-1.

Table 2-1 Summary of Existing Literature

Reference	**Main focus**	**Core method**	**Strength**	**Limitation for this study**
Manunza *et al.* [14]	Real-time VoIP fraud detection	CEP over charging/OCS events	Demonstrates online detection and remediation	Not centred on SIP dialog progression
Seo *et al.* [15]	SIP anomaly detection	Stateful rule tree	Strong protocol awareness and lower overhead	Focus is anomaly/flooding, not toll fraud progression
Chiappetta *et al.* [16]	VoIP user behaviour anomalies	CDR profiling and anomaly analysis	Useful for normal-behaviour baselines	Mainly post-call / CDR-driven

Golait and Hubballi [17]	VoIP anomalous behaviour	Discrete event system modelling	Supports sequence/state-based reasoning	Not targeted specifically at toll fraud
Younes [18]	SIP DDoS detection	Hybrid deep learning + entropy	Strong temporal and protocol-feature modelling	Focused on flooding attacks
Hu *et al.* [19]	Telecom fraud detection	Graph learning	Captures relational fraud behaviour	Not SIP signalling-based
Ren *et al.* [20]	Collaborative fraud detection	Dynamic graph neural networks	Strong for large telecom interaction graphs	Not designed for PBX-integrated early control

2.2.Gap Identification

Although the reviewed literature provides valuable building blocks, several research gaps remain when the specific problem of **early detection of SIP toll fraud** is considered.

The first gap is the absence of an explicit **multi-stage behavioural progression model** for toll fraud. Prior studies recognize that VoIP attacks may involve scanning, authentication abuse, or abnormal calling, but most treat these patterns either separately or as generic anomalies [15], [17], [18], [21]. There is limited work that formally models toll fraud as a sequence of stages such as **reconnaissance ,authentication attack, compromise ,fraud burst**, and then uses those stages to compute an evolving risk level per monitored entity.

The second gap is the weak connection between **SIP signalling-time evidence** and **fraud-specific operational response**. Real-time systems such as Kerberos show that online intervention is possible [14], but they are more event-correlation oriented at the charging level than at the detailed SIP-dialog level. On the other hand, stateful SIP anomaly studies capture protocol behaviour well [15], [17], but they do not generally map that behaviour to toll-fraud-specific actions such as temporary account restriction, route limitation, or call blocking inside PBX operations. This means that a practical gap remains between security analytics and operational PBX control.

The third gap concerns **integration of behavioural baselines with signalling-layer early warning**. CDR- and graph-based studies are highly useful for learning normal calling patterns and suspicious entity relations [16], [19], [20], but they are generally less immediate than signalling-time monitoring. They are therefore stronger for retrospective analysis, fraud scoring, and post-call anomaly interpretation than for detecting fraud before financial damage escalates. What is still missing is an approach that uses CDR-derived behavioural priors to reduce false alarms while keeping the primary detection process close to live SIP events.

The fourth gap lies in the **evaluation criteria used in prior work**. Many studies emphasize classification accuracy, anomaly detection rate, or traffic-level performance [15], [18], [21]. While these are important, operational environments require additional measures such as **false-alert rate, time-to-detect, runtime overhead, and mitigation responsiveness**. These metrics are critical in PBX security because a model that is highly accurate but too slow, too noisy, or

too resource-intensive may not be practical in production. Existing literature does not sufficiently combine algorithmic evaluation with operational feasibility assessment.

The fifth gap is related to **deployment realism**. Very few studies directly compare deployment choices such as embedding the detection mechanism inside the PBX versus integrating it as an external application. Yet this comparison is important for real-world adoption because internal integration may provide lower latency, whereas external integration may offer easier maintenance and broader compatibility. This issue is especially relevant to Asterisk-based or similar PBX environments, where integration can occur through modules, SIP logs, packet capture, AMI, ARI, or CDR feeds.

Based on these gaps, the research need can be stated clearly: there is still limited work that combines **(i)** SIP dialog reconstruction, **(ii)** multi-stage behavioural progression tracking, **(iii)** lightweight machine-learning-based refinement, **(iv)** CDR-informed behavioural baselining, and **(v)** operationally meaningful evaluation within one practical framework for early detection of SIP toll fraud.

Accordingly, this study positions itself to address the following consolidated gap:

Existing research provides separate evidence for online fraud monitoring, stateful SIP inspection, CDR-based behavioural analytics, and intelligent attack classification, but there is no sufficiently integrated and operationally evaluated architecture that detects SIP toll fraud as a multi-stage behavioural progression early enough to support practical PBX mitigation actions.

2.3. Conceptual Framework

The conceptual framework for this study is designed to explain how **early detection of SIP toll fraud** can be achieved by combining signalling-layer evidence, behavioural progression modelling, lightweight machine-learning refinement, and operational mitigation logic within one coherent architecture. The framework is grounded in the main literature streams discussed earlier, particularly real-time fraud monitoring [14], stateful SIP anomaly detection [15], behavioural sequence modelling [17], SIP flooding fingerprint analysis [21], and recent intelligent detection methods for SIP attack traffic [18]. It also aligns closely with the research gap and novelty already defined in the topic assessment form, namely the need for a **lightweight progression-aware architecture** that predicts toll fraud before the fraud call burst occurs.

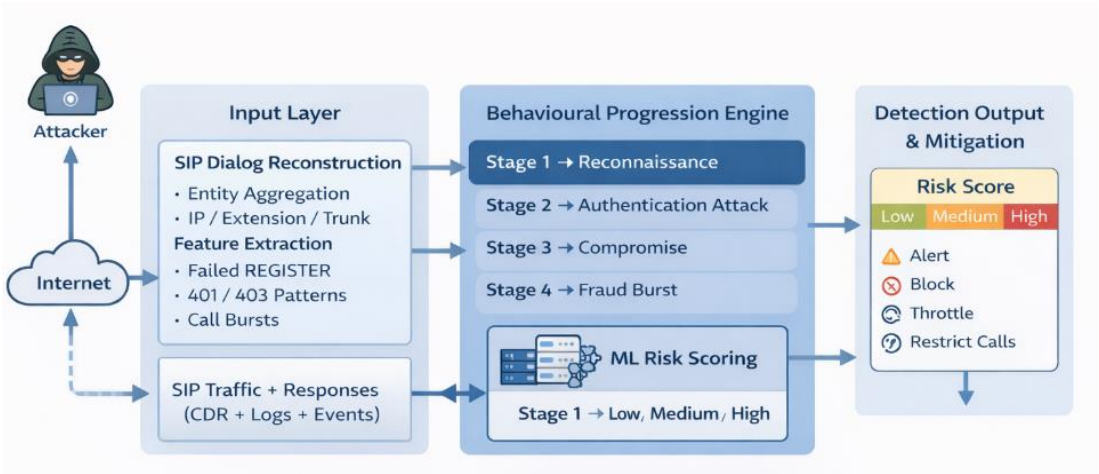
In this study, the framework treats **live SIP signalling behaviour** as the primary source of early attack evidence. SIP requests and responses such as `OPTIONS`, `REGISTER`, `INVITE`, `401 Unauthorized`, `403 Forbidden`, `404 Not Found`, and `200 OK` provide operational indicators of scanning, extension discovery, brute-force registration attempts, compromise, and later fraudulent calling activity [1], [22], [23]. However, isolated messages alone are not sufficient to explain fraud progression. Therefore, the framework introduces a second layer based on **SIP dialog reconstruction and entity-level behavioural aggregation**, where traffic is grouped by

monitored entities such as IP address, extension, trunk, or destination group. From this reconstructed context, the system derives short-window features such as failed-authentication frequency, failure-to-success transition rate, unusual request intensity, inter-arrival irregularity, abnormal destination diversity, and suspicious outbound call bursts [15], [17], [18], [21]. These features form the behavioural evidence used by the next stage of the framework.

At the core of the framework is the **behavioural progression engine**, which models the likely toll fraud lifecycle as a sequence of stages: **reconnaissance → authentication attack → compromise → fraud burst**. This stage-based layer is the main conceptual contribution of the research, because it shifts the focus from isolated event detection toward progression-aware reasoning over time [22], [23]. In order to preserve interpretability while improving detection quality, the framework combines a rule-based progression engine with a **lightweight machine-learning refinement module**. The rule-based component is responsible for interpretable state transitions and explicit attack tracking, while the machine-learning component refines risk estimation by learning subtle relationships among extracted features that may not be fully captured by static rules alone [17], [18]. The combined output is an evolving **risk score** and **stage label** for each monitored entity.

The final layer of the conceptual framework connects analytics to **operational mitigation and evaluation outcomes**. Once the system assigns a sufficient risk level, the framework maps the result to response actions such as alerting, throttling, temporary blocking, route restriction, or account isolation. This makes the framework operationally meaningful rather than purely analytical [14], [22]. The study further includes a deployment dimension, where the same conceptual model is examined under two technical realization options: **an in-process PBX implementation** and **an external detection service** integrated through interfaces such as SIP logs, packet capture, AMI/ARI, and CDR feeds [6], [7]. These deployment choices are not separate concepts outside the framework; instead, they act as practical realization paths that influence latency, maintainability, scalability, and computational overhead. The overall dependent outcome of the framework is therefore **effective early detection of SIP toll fraud**, evaluated using detection accuracy, false-positive behaviour, detection latency, runtime overhead, and mitigation responsiveness.

Table 2-1 Proposed highlevel stage-aware SIP toll fraud detection architecture



2.4.Explanation of the framework

As visualized in the detailed conceptual architecture shown in Figure 2-1, the proposed "Conceptual Framework for Stateful, Adaptive SIP Toll Fraud Early Detection" organizes the analytical process into four distinct, sequential layers. This framework is explicitly designed to shift SIP security from reactive anomaly detection to proactive, state-aware threat management. The flow ensures a structured transition from passive raw data observation to active, automated mitigation outcomes.

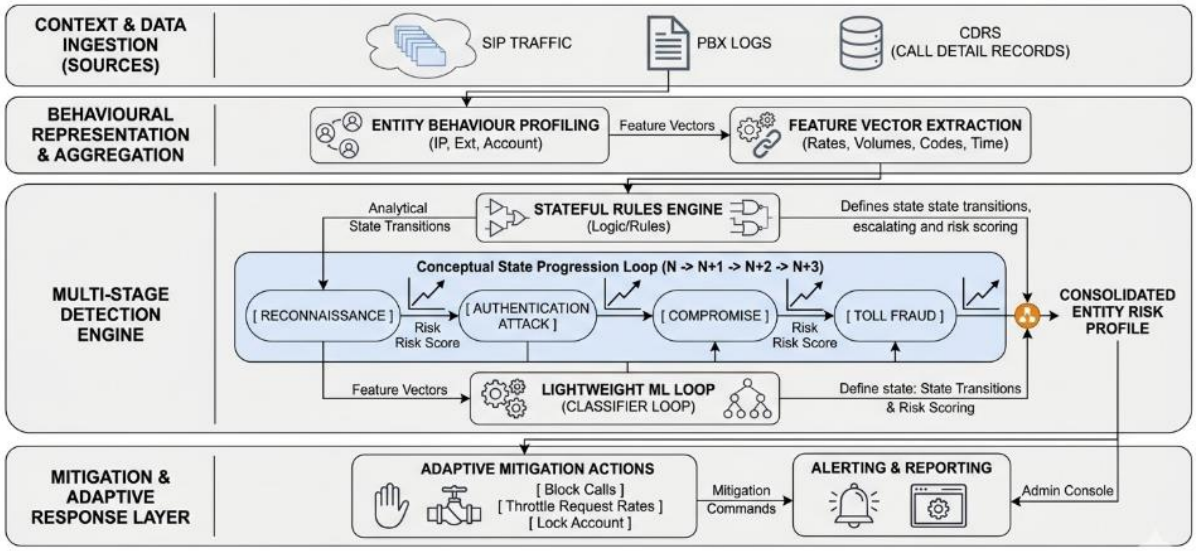


Figure 2-1. Detailed Conceptual Architecture of the Framework

Context & Data Ingestion (Sources)

This foundational layer focuses on aggregating observable evidence from the VoIP environment. To provide a comprehensive security context, the framework must ingest multiple, heterogeneous data types. As shown in Figure 2-1, this layer specifically captures: **SIP Traffic** (the cloud source containing both live signaling signaling packets, requests, responses, IDs, and tags), internal server-generated **PBX Logs** (capturing software event data, such as authentication attempts), and operational **CDRs (Call Detail Records)** (database icons providing metadata on completed calls, durations, and billing destinations). This layer combines real-time technical evidence with broader supporting operational context, providing the essential input volume required to recognize complex, evolving modern fraud patterns.

Behavioural Representation & Aggregation

Raw, siloed data packets are insufficient for detecting multi-stage toll fraud attacks. This layer is responsible for translating individual technical events into a structured entity profile. As depicted in Figure 2-1, this layer performs two critical functions:

- **Entity Behaviour Profiling:** Ingested data is immediately aggregated and associated with unique monitored identities (e.g., an IP address, a specific extension, or an account ID).

- **Feature Vector Extraction:** Once aggregated, meaningful metrics are derived from the raw data streams to create identity-linked **Feature Vectors**. The diagram highlights the specific features extracted: **Rates** (request and traffic velocity), **Volumes** (data transfer quantities), SIP response **Codes** (tracking error trends), and transaction **Time** (duration and overlapping call analysis).

Multi-Stage Detection Engine

This layer represents the core analytical innovation of the proposed research. It receives the Feature Vectors from the previous stage and actively applies a stateful analysis to track the evolution of a potential threat.

The Engine (the central logic core in Figure 2-1) performs continuous analysis using two integrated components:

- **Stateful Rules Engine (Logic/Rules):** This component applies predefined state-transition logic based on observable Feature Vectors.
- **Lightweight ML Loop (Classifier Loop):** As highlighted by the small decision-tree icon, this component is used as a classifier refinement step, validating and optimizing the risk scoring produced by the stateful engine.

These components together feed into the **Conceptual State Progression Loop**. This process determines if an entity is migrating through the logical fraud life cycle stages: [RECONNAISSANCE] → [AUTHENTICATION ATTACK] → [COMPROMISE] → [TOLL FRAUD].Crucially, the **Risk Score** is escalated iteratively as the entity advances across the sequence. The engine combines the stage assessment and classifier refinement to output a consolidated analytical result: a unique **CONSOLIDATED ENTITY RISK PROFILE**.

Mitigation & Adaptive Response Layer

The primary objective of this final layer is to operationalize the consolidated analytical output, allowing the system to react proactively to identified threats well before critical financial damage occurs.

- **Adaptive Mitigation Actions:** The framework can automatically execute granular system-level interventions. As shown in Figure 2-1, these **Mitigation Commands** handle threats directly, including: [Block Calls], [Throttle Request Rates], and [Lock Account].
- **Alerting & Reporting:** Simultaneously, the system generates relevant security alerts for administrators, including Admin Console notifications, email alerts, or SMS messages.

Deployment Realization Options

The same conceptual framework can be implemented in two ways during this research: internally within the PBX stack or externally as an integrated detector. This part is included because implementation choice can directly affect latency, maintainability, and runtime overhead [6], [7], [22].

3. Research Objectives and Research Questions

This section turns the research problem, literature review, and conceptual framework into a practical research plan. Since this study is about building and testing a technical detection system, the main focus is on clear objectives, clear research questions, and a method that can be tested in a realistic VoIP environment.

3.1 Definitions Used in This Proposal

To keep the study clear and consistent, the following terms are used throughout this proposal:

- **Entity**: a monitored subject such as a SIP account or extension, remote IP address, trunk identity, or gateway route.
- **Event**: an observable occurrence within a time window, such as a REGISTER request with a 401 response or multiple INVITE requests within a short period.
- **Stage**: a clear behavioural mode in the attack progression model, namely **Reconnaissance, Authentication Attack, Compromise, Toll Fraud**.
- **Early detection**: generating a **high-risk alert** before, or exactly at the beginning of, the toll-fraud burst stage. This will be measured as the time from a defined onset marker to the first high-risk alert. The onset markers are defined later in the evaluation plan.

3.1. Research Objectives

The main aim of this research is to design, build, and evaluate an adaptive multi-stage behavioural progression architecture for early detection of SIP toll fraud in VoIP environments.

To achieve this aim, the study will address the following objectives:

1. **To identify the main SIP and behavioural indicators** that appear before toll fraud starts.
2. **To develop a stage-aware progression model** that tracks fraud risk for each monitored entity, such as an IP address, extension, trunk, or destination group.
3. **To implement the proposed framework in a VoIP lab environment** and examine both in-process PBX deployment and external detector deployment.
4. **To apply lightweight machine-learning refinement** to improve risk scoring and reduce false alerts.
5. **To evaluate the framework in practical conditions** using normal traffic, simulated attack scenarios, and baseline behaviour from CDR patterns, with measures such as accuracy, false positives, time-to-detect, and overhead, as detailed in Table 3-1.

Table 3-1 Research objectives and practical focus

Objective	Practical focus	Main evaluation points
O1	Identify early fraud indicators	Useful features and clear separation between normal and suspicious behaviour
O2	Develop progression model	Correct stage tracking and risk increase over time
O3	Implement the framework	Deployment feasibility, latency, and maintainability
O4	Add ML refinement	Better precision and fewer false alerts
O5	Evaluate overall performance	Accuracy, speed, overhead, and response quality

3.3 Research Questions and Hypotheses

This study is guided mainly by research questions because the research is focused on designing and testing a technical system. A small set of hypotheses is also included to support comparison during evaluation.

Main research question:

How can an adaptive multi-stage behavioural progression architecture, based on SIP dialog behaviour and feature analysis, detect SIP toll fraud early in practical VoIP environments?

Research questions:

RQ1. Which SIP signalling patterns and behavioural indicators are most useful for identifying early toll fraud progression?

RQ2. Can a stage-aware progression model detect suspicious behaviour earlier than simple request-only or single-event methods?

RQ3. Does lightweight machine-learning refinement improve precision, false-alert control, and overall risk scoring when combined with stage-based tracking?

RQ4. How do the two deployment options—internal PBX integration and external detector integration—compare in terms of latency, overhead, maintainability, and practical use?

Hypotheses:

H1. The proposed multi-stage framework will detect SIP toll fraud earlier than basic baseline methods.

H2. Adding lightweight machine-learning refinement will reduce false alerts and improve overall detection quality compared with a rule-only stage engine.

H3. The in-process deployment option will provide lower detection latency, while the external deployment option will be easier to maintain and still provide acceptable overhead.

4. Research Design and Methodology

This section explains how the proposed framework will be designed, implemented, and evaluated. The methodology is planned to keep the study technically strong, repeatable, and close to real VoIP operating conditions.

4.1 Research Design

This research follows a **design science** approach together with **controlled experimental evaluation**. Design science is suitable because the study aims to build a working technical artefact, which is an early detection framework for SIP toll fraud. Controlled evaluation is also necessary because the framework must be tested under known conditions to check whether it improves early detection in a measurable way.

The study will be carried out in five main phases. First, the problem and system requirements will be refined using the literature review and conceptual framework. Second, the proposed architecture will be defined in terms of data inputs, behavioural features, stage logic, risk scoring, and deployment interfaces. Third, a prototype will be implemented in a VoIP test environment. Fourth, labelled datasets will be created using normal traffic, simulated attack traffic, and sanitized operational records. Finally, the framework will be evaluated using both technical and practical performance measures such as Precision, Recall, F1-score, ROC-AUC, false-alert rate, time-to-detect, and runtime overhead.

This design includes two main comparisons. First, a **rule-based stage model** will be compared with a **stage model supported by lightweight machine learning**. Second, the framework will be tested under **two deployment modes**: an in-process PBX module or patch and an external detector integration. This helps to evaluate not only whether the framework works, but also which implementation approach is more practical.

Testbed and deployment scenarios

The prototype will be built in a lab testbed using an **Asterisk-based PBX environment**, and where useful, **FreeSWITCH** may also be used for comparison. The main tools and interfaces planned for the testbed are shown in Table 4-1.

Table 4-1 Testbed tools and deployment support

Component	Planned use
Asterisk / FreeSWITCH	Lab PBX environment for signalling, calls, and attack simulation
SIPp	Traffic generation for normal and attack scenarios
tcpdump / Wireshark	Packet capture and timing verification
SIP logs	Additional signalling evidence for reconstruction and analysis
AMI / ARI	PBX monitoring, event collection, and control actions
CDR export	Normal behavioural baseline and fraud context

Two deployment scenarios will be examined during implementation:

- **Scenario A: In-process PBX module or patch** – the detection logic runs inside the PBX environment for lower latency.

- **Scenario B: External integrated detector** – a separate application receives signalling and PBX events, performs analysis, and sends actions back to the PBX.

4.2 Sampling Strategy

This study does not use human participants. Instead, the sampling unit is technical activity generated inside the VoIP environment. This includes SIP messages, reconstructed dialogs, monitored entities, and related CDR records. Therefore, the study will use a **purposive and scenario-based sampling strategy**.

The collected traffic will be grouped into five main categories. The first category is **normal traffic**, such as valid registrations, successful calls, and ordinary destination patterns. The second category is **reconnaissance activity**, such as extension discovery, probing, and invalid-user requests. The third category is **authentication attack activity**, such as repeated failed registrations and unauthorized attempts. The fourth category is **compromise and fraud activity**, such as rapid failure-to-success transitions, unusual destinations, and abnormal outbound bursts. The fifth category is **benign burst activity**, such as a valid call-centre traffic increase, which is important for testing whether the framework can avoid false high-risk alerts.

To improve realism, the study will combine **controlled synthetic traffic** with **sanitized historical CDR patterns** where available. If real CDR data is not available, a realistic baseline will be generated using practical calling patterns in the lab environment. Attack scenarios will also be repeated under low, medium, and high intensity so that the framework can be tested under different conditions, as shown in Table 4-2.

Table 4-2 Traffic categories used in the study

Traffic category	Typical examples	Purpose
Normal traffic	Valid registration, normal calls, usual retry patterns	Build baseline and measure false alerts
Reconnaissance	Enumeration, probing, invalid-user discovery	Test early-stage detection
Authentication attack	Repeated failures, unauthorized access attempts	Test progression before compromise
Compromise / fraud	Rapid failure-to-success patterns, unusual destinations, outbound bursts	Test fraud-stage detection and response
Benign burst	Legitimate high call load without compromise	Test robustness and reduce false high-risk alerts

4.3 Data Collection Methods

Data will be collected in a controlled **VoIP testbed** with extensions, trunks, routing rules, and logging functions similar to a real PBX environment. The study will use multiple evidence sources, including SIP messages, response codes, dialog identifiers such as Call-ID and tags, operational events, packet-capture traces, and CDR records.

The main data sources, as shown in Table 4-3, will include:

1. **Labelled lab SIP traffic** generated through normal calling and registration patterns together with simulated attack stages.

2. **Sanitized real CDR data**, where available, to build a more realistic normal baseline.
3. **PBX event data** collected through interfaces such as AMI and ARI.
4. **Packet captures and SIP logs** used for timing checks, verification, and feature extraction.

Normal datasets will be produced by running ordinary registration and calling scenarios over a defined observation period. Fraud-related datasets will be produced by injecting controlled attack scenarios such as reconnaissance, password guessing, account compromise, and fraudulent outbound calling. Since these scenarios are created in a controlled environment, the ground truth will be known and can be used for training and testing.

Each event will be linked to one or more monitored entities such as source IP, extension, destination number, or trunk. During preprocessing, the raw data will be synchronized and normalized into one analysis dataset. Labels will then be assigned to represent normal activity, reconnaissance, authentication attack, compromise, or fraud burst.

Table 4-3 Main data sources and purpose

Data source	Examples of captured data	Use in the study
SIP signalling records	Method, response code, Call-ID, tags, timestamps, source and destination fields	Dialog reconstruction and feature extraction
Operational event streams	AMI/ARI events, registration changes, channel state changes	Context support and deployment testing
Packet-capture traces	Timing, request order, retransmissions	Verification and timing analysis
Call Detail Records	Destination, duration, start/end time, trunk usage	Behavioural baselines and fraud context

4.4 Data Analysis Techniques

The first stage of analysis will focus on **preprocessing and feature generation**. SIP records will be cleaned, time-normalized, and reconstructed into transactions and dialogs using SIP identifiers and protocol relationships. The events will then be grouped into observation windows for each monitored entity.

A default sliding window of **60 seconds** with a **5-second step** will be used at the beginning, and older evidence may be reduced using decay logic. From these windows, the study will extract features such as failed REGISTER attempts, 401/403/404 response patterns, failure-to-success transitions, request intensity, response-code distribution, inter-arrival irregularity, destination diversity, concurrent session count, off-hours activity, and outbound burst behaviour, as shown in Table 4-4

Table 4-4 Main feature groups used in the study

Feature group	Example features	Purpose
Rate and volume	REGISTER rate, INVITE rate, unique targets	Detect recon and brute-force increase

Response-code behaviour	Counts or ratios of 200, 401, 403, 404, 480	Capture probing, failure, and compromise signs
Timing and burstiness	Inter-arrival gap, retry density, burst factor	Detect abnormal timing patterns
Dialog integrity	Call setup completion rate, early BYE ratio	Separate real calls from abnormal signalling
Entity drift	New IP, user-agent change, unusual time-of-day change	Support compromise suspicion
CDR-based behaviour	Typical calls per hour, normal destinations, duration pattern	Reduce false alerts using normal behaviour context

The second stage will focus on **progression-aware detection**. A rule-based stage engine will map behavioural evidence into the expected fraud lifecycle: reconnaissance, authentication attack, compromise, and fraud burst. On top of this, a lightweight supervised learning model, such as logistic regression or a small tree-based classifier, will be used to refine the risk score.

The framework will be compared against two simple baselines:

- **Baseline A:** request-only detection based on rates and target spread.
- **Baseline B:** request-response correlated detection using response-code ratios and timing relationships.

A final risk score will then be used to support response actions. In simple terms, low risk will only be logged, medium risk will trigger an alert and limited restriction, and high risk will trigger active mitigation such as temporary blocking, route restriction, or account control.

The third stage will focus on **performance evaluation**. The outputs will be assessed using confusion matrices and metrics such as Precision, Recall, F1-score, ROC-AUC, and false-alert rate. Early detection ability will be measured using **time-to-detect**, which is the time from a defined onset marker to the first high-risk alert. Operational feasibility will be examined using CPU usage, memory usage, detection processing time, and possible PBX call setup impact under each deployment option.

Finally, the proposed framework will be compared with the basic baseline detectors and also with the rule-only version. Repeated test runs will be summarized using average values and standard deviations. Where suitable, simple paired statistical tests will be used to check whether the observed improvements are meaningful.

5. Significance and Expected Contributions

This research addresses SIP toll fraud as both a critical technical vulnerability and a significant business risk. By focusing on early detection before the costly "fraud burst" occurs, the study provides value on multiple fronts.

5.1 Significance of the Study

Academic Value: Moves beyond single-event anomaly detection by creating a unified framework that combines SIP dialog behavior, multi-stage progression tracking, and lightweight machine learning.

Practical Value: Delivers an early-warning system that alerts PBX administrators and VoIP operators *before* financial damage escalates, unlike traditional post-event billing analysis.

Deployment Realism: Tests the model in realistic conditions by comparing an in-process PBX deployment against an external integrated detector to ensure maintainability and low latency.

Operational Focus: Evaluates success using practical metrics—such as false alerts, time-to-detect, and runtime overhead—rather than relying solely on theoretical detection accuracy.

5.2 Expected Contributions

The expected contributions of this study are as follows, and summarized in Table 5-1:

Table 5-1 Expected contributions of the study

Contribution area	Expected contribution
Conceptual contribution	A clear multi-stage model for SIP toll fraud progression
Methodological contribution	A combined approach using signalling, behaviour, and lightweight ML
Technical contribution	A prototype framework that can be tested in practical VoIP settings
Evaluation contribution	A set of operational metrics for early fraud detection assessment
Practical contribution	Guidance for real deployment in PBX and VoIP environments

6. Ethical Considerations

This research mainly uses technical data collected from a controlled VoIP testbed. It does not involve human subjects, medical data, or direct personal interviews. However, ethical considerations are still important because the work deals with security monitoring, simulated attack behaviour, and possibly sanitized operational data.

6.1 Ethical Guidelines

The following ethical guidelines will be followed during the study:

- **Controlled and authorized environment** – all experiments will be carried out only in an approved lab or test environment. No attack simulation will be performed against live third-party systems.
- **Use of sanitized data** – if real CDR or operational data is used, it will be anonymized or sanitized before analysis. Sensitive identifiers such as phone numbers, usernames, IP addresses, or organization details will be masked where needed.
- **Data minimization** – only the data needed for the research objectives will be collected and stored.
- **Secure data handling** – datasets, captures, logs, and analysis outputs will be stored securely and accessed only for academic research purposes.
- **No harmful deployment** – the framework will be developed for defensive detection and mitigation. The research will not provide tools or instructions intended to support misuse.
- **Research integrity** – all sources will be properly referenced, and the results will be reported honestly, including limitations, negative findings, and unexpected outcomes.
- **Supervisor and institutional compliance** – the work will follow university research guidelines and any additional instructions provided by the supervisor or department.

7. Timeline and Work Plan

This section presents the expected timeline for completing the study. The work plan is based on the current progress of the proposal. Up to this stage, the topic definition, literature review, gap analysis, conceptual framework, and the main methodology design have already been developed. The next stages focus on prototype building, dataset preparation, testing, evaluation, and final writing.

7.1. Research Timeline

The study is expected to be completed through the following phases, as shown in Table 7-1.

Table 7-1 Research timeline and work plan

Phase	Main activities	Status
Phase 1	Topic selection, problem definition, proposal planning	Completed
Phase 2	Literature review, gap identification, conceptual framework	Completed
Phase 3	Finalize objectives, questions, and methodology	In progress
Phase 4	Design system architecture and feature plan	Planned
Phase 5	Build testbed and implement prototype	Planned
Phase 6	Generate normal and attack traffic datasets	Planned
Phase 7	Run experiments and collect results	Planned
Phase 8	Evaluate model performance and compare deployment options	Planned
Phase 9	Write final report, revise, and prepare presentation	Planned

A suggested month-based timeline is provided in Table 7-2.

Table 7-2 Suggested month-based timeline

Research activity / Milestone	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Detailed architecture and experiment design		■	■								
Progress Presentation 1 – **4 May**				■							
Prepare lab setup and data collection pipeline				■	■						
Implement stage-based detection logic					■	■					
Implement ML refinement and baseline models						■	■				
Journal Submission to Supervisor – **15 Aug**							■				
Generate datasets and run initial experiments							■	■			
Draft Thesis Submission – **14 Sep**								■			
Progress Presentation 2 – **15 Sep**								■			
Tune features, windows, and risk thresholds								■			
Compare deployment options and measure overhead								■	■		
Supervisor Recommendation Form – **15 Oct**									■		

Final analysis and write results chapters											
Thesis Submission for Examiners – **3 Nov**											
Final editing and proofreading											
Final Thesis Presentation – **1 Dec**											
Final Thesis deposit in library/repositories – **31 Dec**											

8. Resources and Budget

This section identifies the main resources needed for the study. The proposed research is expected to require more technical resources than financial resources because most of the development can be carried out using open-source tools and an internal lab setup.

8.1. Resource Requirements

The main resources required for this study are listed below and summarized in Table 8-1.

Software resources

- Asterisk and, if needed, FreeSWITCH for the VoIP test environment
- SIPp for traffic generation
- Wireshark and tcpdump for packet capture and protocol analysis
- Python and related libraries for data processing, feature extraction, and machine learning
- Database or structured storage for logs, events, and CDR-based analysis

Hardware and infrastructure resources

- A lab server, virtual machine, or workstation to host the PBX environment
- A separate machine or virtual environment for the external detector
- Network access for controlled SIP traffic generation and packet capture
- Sufficient storage for logs, captures, datasets, and experiment outputs

Data resources

- Lab-generated normal SIP traffic
- Simulated SIP attack traffic
- Sanitized historical CDR patterns, if available

Table 8-1 Main resource requirements

Resource category	Requirement	Availability
PBX platform	Asterisk / optional FreeSWITCH	Available / planned
Traffic generator	SIPp	Available
Monitoring tools	Wireshark, tcpdump, SIP logs	Available
Development tools	Python, ML libraries, scripting tools	Available
Data storage	Local or virtual storage for logs and datasets	Planned
Test environment	VM or lab server	Planned
Reference material	Journal articles, documentation, proposal drafts	Available

Budget note

The expected direct financial cost of this study is relatively low because the main tools are open source and the work is expected to use an academic or internal lab environment. If extra cost occurs, it is most likely to be related to computing resources, storage, or test environment setup. Therefore, the study is considered feasible with a low-budget research setup.

9. References

[1] J. Rosenberg et al., “SIP: Session Initiation Protocol,” RFC 3261, IETF, Jun. 2002, doi: 10.17487/RFC3261.

[2] L. Manunza, S. Marseglia, and S. P. Romano, “Kerberos: A real-time fraud detection system for IMS-enabled VoIP networks,” Journal of Network and Computer Applications, vol. 80, pp. 22–34, Feb. 2017, doi: 10.1016/j.jnca.2016.12.018.

[3] D. Seo, H. Lee, and E. Nuwere, “SIPAD: SIP–VoIP anomaly detection using a stateful rule tree,” Computer Communications, vol. 36, no. 5, pp. 562–574, Mar. 2013, doi: 10.1016/j.comcom.2012.12.004.

[4] D. Golait and N. Hubballi, “Detecting anomalous behavior in VoIP systems: A discrete event system modeling,” IEEE Transactions on Information Forensics and Security, vol. 12, no. 3, pp. 730–745, Mar. 2017, doi: 10.1109/TIFS.2016.2632071.

[5] O. S. Younes, “A hybrid deep learning model for detecting DDoS flooding attacks in SIP-based systems,” Computer Networks, vol. 240, Art. no. 110146, Feb. 2024, doi: 10.1016/j.comnet.2023.110146.

[6] “Asterisk REST Interface (ARI) Overview,” Asterisk Documentation. [Online]. Available through official Asterisk documentation.

[7] “Call Detail Record (CDR) Drivers,” Asterisk Documentation. [Online]. Available through official Asterisk documentation

[8] H. Sengar, D. Wijesekera, H. Wang, and S. Jajodia, “VoIP intrusion detection through interacting protocol state machines,” in *Proc. International Conference on Dependable Systems and Networks (DSN’06)*, 2006.

[9] T. Kapourniotis, T. Dagiuklas, C. G. Polyzos, and P. Alefragkis, “Scam and fraud detection in VoIP networks: Analysis and countermeasures using user profiling,” in *2011 50th FITCE Congress: ICT: Bridging an Ever Shifting Digital Divide*, 2011, pp. 1–5, doi: 10.1109/FITCE.2011.6133427.

[10] T. Cejka, V. Bartos, L. Truxa, and H. Kubatova, “Using application-aware flow monitoring for SIP fraud detection,” in *Intelligent Mechanisms for Network Configuration and Security*, vol. 9122, Cham: Springer, 2015, pp. 87–99, doi: 10.1007/978-3-319-20034-7_10.

[11] W. Nazih, W. S. Elkilani, H. Dhahri, and T. Abdelkader, “Survey of countering DoS/DDoS attacks on SIP based VoIP networks,” *Electronics*, vol. 9, no. 11, Art. no. 1827, 2020, doi: 10.3390/electronics9111827.

[12] “Overview,” *Asterisk Documentation*, Reporting. [Online]. Available: https://docs.asterisk.org/Configuration/Reporting/

[13] “Overview,” *Asterisk Documentation*, Asterisk REST Interface (ARI). [Online]. Available: https://docs.asterisk.org/Configuration/Interfaces/Asterisk-REST-Interface-ARI/

[14] L. Manunza, S. Marseglia, and S. P. Romano, “Kerberos: A real-time fraud detection system for IMS-enabled VoIP networks,” *Journal of Network and Computer Applications*, vol. 80, pp. 22–34, Feb. 2017, doi: 10.1016/j.jnca.2016.12.018.

[15] D. Seo, H. Lee, and E. Nuwere, “SIPAD: SIP–VoIP anomaly detection using a stateful rule tree,” *Computer Communications*, vol. 36, no. 5, pp. 562–574, Mar. 2013, doi: 10.1016/j.comcom.2012.12.004.

[16] S. Chiappetta, C. Mazzariello, R. Presta, and S. P. Romano, “An anomaly-based approach to the analysis of the social behavior of VoIP users,” *Computer Networks*, vol. 57, no. 6, pp. 1545–1559, Apr. 2013, doi: 10.1016/j.comnet.2013.02.009.

[17] D. Golait and N. Hubballi, “Detecting anomalous behavior in VoIP systems: A discrete event system modeling,” *IEEE Transactions on Information Forensics and Security*, vol. 12, no. 3, pp. 730–745, Mar. 2017, doi: 10.1109/TIFS.2016.2632071.

[18] O. S. Younes, “A hybrid deep learning model for detecting DDoS flooding attacks in SIP-based systems,” *Computer Networks*, vol. 240, Art. no. 110146, Feb. 2024, doi: 10.1016/j.comnet.2023.110146.

[19] X. Hu *et al*., “BTG: A bridge to graph machine learning in telecommunications fraud detection,” *Future Generation Computer Systems*, vol. 137, pp. 274–287, Dec. 2022, doi: 10.1016/j.future.2022.07.020.

[20] L. Ren *et al*., “Dynamic graph neural network-based fraud detectors against collaborative fraudsters,” *Knowledge-Based Systems*, vol. 278, Art. no. 110888, Oct. 2023, doi: 10.1016/j.knosys.2023.110888.

[21] K. Dassouki, H. Safa, M. Nassar, and A. Hijazi, "Protecting from cloud-based SIP flooding attacks by leveraging temporal and structural fingerprints," Computers & Security, vol. 70, pp. 618–633, 2017, doi: 10.1016/j.cose.2017.08.003.

[22] A. Aziz, Real-time Detection & Mitigation of SIP-based VoIP Toll Fraud Attacks, Ph.D. dissertation, University of Duisburg-Essen, Essen, Germany, 2024.

[23] A. Aziz, D. Hoffstadt, E. Rathgeb, and T. Dreibholz, "A Distributed Infrastructure to Analyse SIP Attacks in the Internet," in IFIP Networking Conference, Trondheim, Norway, 2014.

10. Appendices

DECLARATION

I declare that this is my own research proposal, and this proposal does not incorporate without acknowledgement any material previously published, submitted for a Degree or Diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.



Signature: Date: 24/03/2026
K.M.U.P Kosgoda

I have read the proposal, and it is in accordance with the approved university proposal outline. I am willing to supervise the research work of the above candidate on the proposed area.

Signature: Date:
Dr. Nimal Rathnayake[